QP Code:25901214

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2026

FOURTH SEMESTER

Discipline Specific Elective (DSE)- MG4DSEECO200

EXPLORING ECONOMIC DATA ANALYTICAL TOOLS

(2024 ADMISSION ONWARDS)

SCHEME OF VALUATION

Duration: 1 Hours 30 Minutes **Maximum Marks:50**

Part A – Multiple Choice Questions **(1 × 15 = 15 marks)**

1. b) Statistics
2. a) Analyze > Descriptive Statistics
3. b) Interactive exploration
4. a) Right-click series > View > Histogram
5. b) .gdt
6. b) Free, cross-platform
7. a) Large-scale survey analysis
8. b) Microeconometrics and panel data
9. a) View libraries and datasets
10. a) Thousands of specialized packages
11. a) Interactive business insights
12. a) Power Query Editor
13. b) Ross Ihaka and Robert Gentleman
14. b) 4 panes
15. b) 2015

Part B – Short Answer Questions (Any 10 × 2 = 20 marks)

16. Graphical tools help in better understanding of economic data by visually presenting trends and patterns. They simplify complex numerical data and improve interpretation for decision-making.
17. SPSS has limitations such as restricted flexibility in advanced econometric modelling. It is also costly and less suitable for handling large-scale or highly complex datasets.
18. SPSS is widely used for descriptive statistical analysis such as mean and variance. It is also used for regression analysis and survey data interpretation in social sciences.
19. EVIEWS simplifies univariate analysis by providing easy computation of descriptive statistics. It also offers graphical tools like histograms and line graphs for better data visualization.
20. EVIEWS supports data operations like sorting and transforming variables. It also

allows creation of new variables and data cleaning for analysis.

21. GRETL is user-friendly and suitable for beginners in economic modelling. It supports basic statistical functions and is free and open-source.
22. SAS is highly efficient in handling large datasets in surveys. It also provides strong data management and advanced statistical capabilities.
23. Common STATA commands include “generate” for creating variables. “replace” is used to modify existing data values.
24. SAS uses commands like DATA and INFILE for importing data. The PROC IMPORT statement is commonly used to load external datasets into SAS.
25. R packages such as “dplyr” help in data manipulation. “ggplot2” is useful for data visualization in economic analysis.
26. Measures of dispersion in R include standard deviation and variance. Range and interquartile range can also be calculated.
27. Slicers in Power BI allow users to filter data interactively. They help in focusing on specific variables and improve dashboard usability.
28. Power BI supports univariate visualization through charts like bar charts and histograms. It enables easy drag-and-drop features for quick analysis.

Part C – Short Essay Questions **(Any 3 × 5 = 15 marks)**

29. SPSS provides simple steps to create charts for economic data visualization. Users can access the Graphs menu or Chart Builder to select chart types such as histograms, pie charts, and bar charts. Data variables are assigned to axes, and charts are generated automatically for analysis.

    Customization options in SPSS allow users to modify titles, labels, colours, and scales. These visual tools help economists identify patterns, frequency distributions, and comparisons, making data interpretation clearer and more effective.

30. SPSS is widely preferred due to its user-friendly interface, which allows economists to perform analysis without advanced programming skills. Menu-driven commands and dialog boxes simplify statistical operations, making it suitable for beginners and researchers.

    In addition, SPSS has a rich statistical library that includes descriptive statistics, regression, and hypothesis testing tools. This combination enhances efficiency and accuracy in data analysis, making it a valuable tool in economic research.

31. EVIEWS is widely used for time series and econometric analysis in economics. It

allows users to input time-based data and perform operations such as trend analysis, forecasting, and regression modelling.

In practical applications, EVIEWS is used to study GDP growth, inflation trends, and financial market movements. Its built-in econometric tools help researchers estimate models and test economic theories effectively.

32. STATA is a powerful tool for data management and statistical analysis. It provides efficient commands for handling large datasets and supports various econometric techniques, including regression and panel data analysis.

    However, STATA has limitations such as requiring command knowledge for advanced use and being costly. Despite this, its speed, reliability, and wide application in research make it highly valuable for economists.

33. R uses a flexible syntax that allows users to perform complex data manipulation efficiently. Functions and packages enable operations such as data cleaning, transformation, and statistical modelling.

    Moreover, R supports advanced statistical techniques and graphical representation through packages like ggplot2. Its open-source nature and extensibility make it highly suitable for complex economic data analysis.